



Full Length Article

Nanocrystalline Sm^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ garnets: An intense orange-reddish luminescent material for white light emitting devicesP. Haritha^a, C.S. Dwaraka Viswanath^b, K. Linganna^a, P. Babu^c, C.K. Jayasankar^b, V. Lavín^d, V. Venkatramu^{a,*}^a Department of Physics, Yogi Vemana University, Kadapa 516003, India^b Department of Physics, Sri Venkateswara University, Tirupati 517502, India^c Department of Physics, Government Degree College, Satyavedu 517588, India^d Departamento de Física, IUEA, and MALTA Consolider Team, Universidad de La Laguna, Apdo. 456, 38200 San Cristóbal de La Laguna, Santa Cruz de Tenerife, Spain

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ABSTRACT

Nanocrystalline lutetium gallium garnets $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ doped with various concentrations of Sm^{3+} ions have been prepared by a low-cost and environmentally benign sol–gel method. The nanocrystalline structure of the $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ garnets has been identified by X-ray powder diffraction whereas their optical properties have been investigated through laser-excited spectroscopic techniques. The diffuse reflectance spectrum of Sm^{3+} ions in $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ nano-garnets has been measured for predicting its partial energy level structure. The luminescence spectra of Sm^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ nano-garnets have been measured under both 405 and 473 nm laser excitations as a function of Sm^{3+} ion concentration. The nano-garnets exhibit intense orange-reddish luminescence at 615 nm attributed to the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition. The optimum molar concentration of Sm^{3+} ions have been found to be about 3.0 mol% with critical energy transfer distance of 16.8 Å. The luminescence decay curves of the $^4\text{G}_{5/2}$ level of Sm^{3+} ions measured under 560 nm resonant excitation have been found to be non-exponential. The non-exponential nature of the decay curves has been analyzed by Inokuti–Hirayama model and found that the dominant mechanism of interaction between active ions is of dipole–dipole type. All these results indicate that the present nano-garnets could be useful for white light emitting devices.

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1. Introduction

Recently, trivalent rare-earth (RE^{3+}) ions activated oxide-based nano-crystals are attracted immense attention for the design and development of white light-emitting devices (WLEDs) for solid-state lighting applications as they have long lifetime, high luminescence efficiency, low power consumption, high chemical stability and environment friendly characteristics [1–3]. The WLEDs were fabricated by combining the near-UV or UV LED with the red, green, and blue (RGB) tricolor phosphors. Usually, white light has been generated with the combination of $\text{ZnS}:\text{Cu}^+$, Al^{3+} (green), $\text{Y}_2\text{O}_2\text{S}:\text{Eu}^{3+}$ (red), and $\text{BaMgAl}_{10}\text{O}_{17}:\text{Eu}^{2+}$ (blue) luminescent materials [4]. However, the efficiency of red luminescence from $\text{Y}_2\text{O}_2\text{S}:\text{Eu}^{3+}$ is less than that of green and blue phosphors, besides its poor chemical stability at high temperature [4]. In this direction, there is a need for rigorous search for developing red

emitting phosphors with high luminescence intensity, good chemical stability and strong absorption in the near-UV region. In this context, selection of host material and suitable dopant with optimum concentration is a challenging task because the luminescent efficiency mostly depends on the physical and chemical properties of the host as well as dopant concentration.

Garnets are good host matrices for RE^{3+} ions as they exhibit interesting properties like high chemical stability, high thermal conductivity, hardness and show intense luminescence useful for WLEDs application when activated with RE^{3+} ions [2,5]. The nanocrystalline garnets or nano-garnets are of immense interest to study since their structural properties are similar to their bulk counterparts [6]. Moreover, the nano-garnets are synthesized by a simple, low-cost, low-temperature and environment friendly sol–gel technique whereas their bulk counterparts are prepared by high temperature and high-cost crystal growth techniques with consumption of relatively more time. Rao et al. [7] synthesised the un-doped and Sm^{3+} -doped $\text{Ca}_3\text{Y}_2\text{Si}_3\text{O}_{12}$ phosphor for intense red-orange emission, but the material did not get crystallize in the garnet structure and it is partly distorted from the structure.

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Gallium (Ga^{3+}) based garnets are interesting because of their larger unit cell volume, high refractive index and high solubility of RE^{3+} ions, i.e., low concentration quenching, larger stimulated emission cross-section, lower pump thresholds and higher luminescence efficiency [8,9]. Furthermore, lutetium gallium based $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ (LuGG) garnets have been attracted since luminescence intensities of Lu^{3+} -based compounds are significantly higher than corresponding Yttrium (Y^{3+}) or Gadolinium (Gd^{3+}) based compounds [10,11].

Among RE^{3+} ions, Sm^{3+} ion is recognized as an excellent activator in many orange-reddish phosphors since its luminescent $^4\text{G}_{5/2}$ level possesses higher lifetime of the order of milliseconds, resulting high quantum efficiency and intense luminescence in the visible and near-infrared (NIR) regions due to its $^4\text{G}_{5/2} \rightarrow ^6\text{H}_J$ ($J = 5/2-13/2$) transitions. The emission of Sm^{3+} ions associated with the intra-4f shell transition is very efficient and its luminescent $^4\text{G}_{5/2}$ level exhibits long lifetime at low concentrations. Hence, there is an immense attention in Sm^{3+} ion as activator in a variety of materials for orange-reddish phosphors as well as lasers.

The inorganic phosphors doped with Sm^{3+} ions have been investigated for phosphor based WLEDs [12,13]. In this direction, considerable work has been carried out on the Sm^{3+} ions doped hosts such as $\text{Ca}_3\text{Y}_2\text{Si}_3\text{O}_{12}$ [7], $\text{Na}_2\text{CaSn}_2\text{Ge}_3\text{O}_{12}$ [13], Ca_2SnO_4 [14], and CaTiO_3 [15], for orange-reddish emitting and long persistent phosphorescence. Recently, Sm^{3+} -doped Ca_2SnO_4 has shown long lasting phosphorescence for more than 7 h in the orange-reddish region which can find applications in lighting, displays or detection of high energy rays [14]. Moreover, Sm^{3+} ions also find applications as optical temperature sensors to measure high temperatures, typically higher than 800 K. Since, the intensity ratio of $^4\text{F}_{3/2} \rightarrow ^6\text{H}_{5/2}$ to $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{5/2}$ transitions of Sm^{3+} -doped GdVO_4 phosphor was found to be linear in the temperature range of 293–823 K and finds application as temperature sensor [16].

In this work, lutetium gallium $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ nano-garnets doped with various concentrations of Sm^{3+} ions have been prepared and studied their structural and luminescence properties. The structure of the nano-garnets has been identified by X-ray powder diffraction and their optical properties have been characterized through spectroscopic techniques. The concentration dependent orange-reddish luminescence properties and multipolar interactions among the optically active Sm^{3+} ions in the LuGG nano-garnets have been studied in detail to know the optimum concentration to exploit them for the development of WLEDs.

2. Experimental details

A simple and environment-friendly sol-gel method was used to prepare powder nano-garnets of composition $\text{Lu}_{3(1-x)}\text{Sm}_{3x}\text{Ga}_5\text{O}_{12}$, where $x = 0.01, 0.03, 0.05$ and 0.1 , and labeled as LuGG1Sm, LuGG3Sm, LuGG5Sm, and LuGG10Sm, respectively [17]. X-Ray diffraction (XRD) patterns of the synthesized nano-garnets were measured using the $\text{CuK}\alpha_1$ (1.5406 Å) radiation with a step size of 0.02° (RIGAKU; Miniflex-600). The particle size distribution was carried out using a Zetasizer Nano-S90 (Malvern Instruments, USA) by dynamic light scattering (DLS) technique. The diffuse reflectance spectrum in UV-visible-NIR region was measured with a spectrophotometer (Agilent Technologies Cary5000). The luminescence spectra in the range of 500–850 nm were measured by exciting the samples with diode lasers either at 405 or 473 nm. These emissions were focused with a convergent lens onto a fiber coupled 0.303 m single grating spectrograph (Andor Shamrock SR-303i-B) and then detected with a cooled CCD detector (Newton DU920N). The photoluminescence excitation (PLE) spectrum was recorded by Edinburgh FLS980 spectrometer with 450 W xenon arc lamp (cw) as a light source. The luminescence decay curves for

the $^4\text{G}_{5/2}$ level of Sm^{3+} ions were measured under 560 nm xenon flash lamp (60 W) with pulse width of 1–2 μs and pulse repetition rate between 0.1 Hz and 100 Hz. Single grating monochromator with focal length of 0.3 m and cooled photomultiplier tube (PMT, Hamamatsu R928P) as detector was used. All the measurements were carried out at room temperature and corrected for instrument response.

3. Results and discussion

The XRD spectra of Sm^{3+} -doped LuGG nano-garnets were measured in the range of $15-80^\circ$ and are shown in Fig. 1(A). The International Center for Diffraction Data (ICDD) was used to identify the crystallization phases. The diffraction peaks shown in Fig. 1(A) coincide well with the standard JCPDS card 13-0400 and well-indexed to a space group Ia-3d, No. 230. The vertical lines indicate the Bragg reflections and it confirms the formation of single phase cubic Sm^{3+} -doped LuGG nano-garnets, in which the Sm^{3+} ions are assumed to occupy the Lu^{3+} dodecahedral sites. The average crystallite size of the garnets was estimated by using Debye-Scherrer's equation [18] and was found to be around

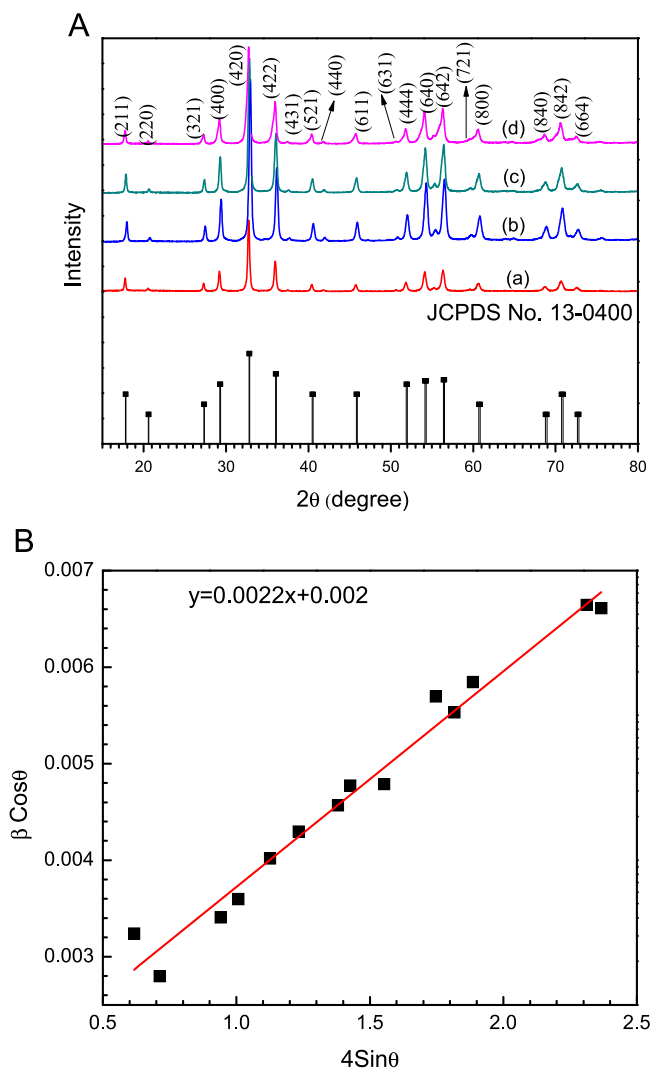


Fig. 1. (A) XRD spectra of (a) 1.0 mol%, (b) 3.0 mol%, (c) 5.0 mol% and (d) 10.0 mol% of Sm^{3+} -doped LuGG nano-garnets. The vertical marks are the allowed Bragg reflections for this material in the Ia-3d (No. 230) space group. (B) The Williamson-Hall Plot of $4\sin \theta$ Vs $\beta \cos \theta$ of 1.0 mol% Sm^{3+} -doped LuGG nano-garnets.

Table 1Lattice parameter, crystallite size, and lattice strain for the Sm^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ nano-garnets along with other nanocrystalline materials.

S. no.	Sample	Lattice parameter (Å)	Crystallite size (nm)		Lattice strain (%)	Ref.
			Debye-Scherrer	Williamson-Hall		
1	LuGG1Sm	$a = 12.2164$	37	69	0.22	Present
2	LuGG3Sm	$a = 12.1694$	32	59	0.34	Present
3	LuGG5Sm	$a = 12.2104$	33	68	0.35	Present
4	LuGG10Sm	$a = 12.2403$	26	43	0.39	Present
5	ZnO	$a = 3.2491$ $b = 5.2063$	27	35	0.13	20
6	$\text{Fe}_2\text{Mn}_2\text{Ni}_{0.5}\text{Zn}_{1.5}\text{O}_9$	$a = 8.3716$ $b = 5.9194$ $c = 3.0457$	17	28	0.40	21

32 nm. However, the defects on the surface of the nano-garnets may cause lattice strain, which leads to broadening of diffraction peaks. Two main factors play a crucial role in peak broadening that are crystallite size and lattice strain. A Williamson–Hall (W–H) method was employed for estimating the lattice strain besides crystallite size using the following equation [19],

$$\beta \cdot \cos \theta = 4 \cdot e \cdot \sin \theta + \frac{k\lambda}{L} \quad (1)$$

where β is the full width at half maximum (FWHM); θ is the diffraction angle; k is the shape factor (0.89); λ is the wavelength of X-rays used; L is the crystallite size; and e is the lattice strain.

A plot between $\beta \cdot \cos \theta$ and $4 \cdot \sin \theta$ is drawn and as shown in Fig. 1(B) for the evaluation of lattice strain and crystallite size. The lattice strain 'e' can be obtained from slope of the line and crystallite size L can be obtained from the intercept. Therefore, the average crystallite size and the strain from the W–H fit were found to be 60 nm and 0.32%, respectively, for Sm^{3+} -doped LuGG nano-garnets.

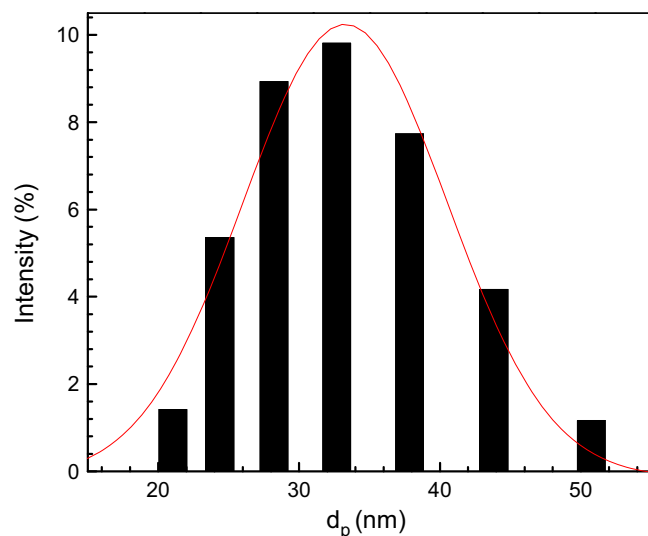
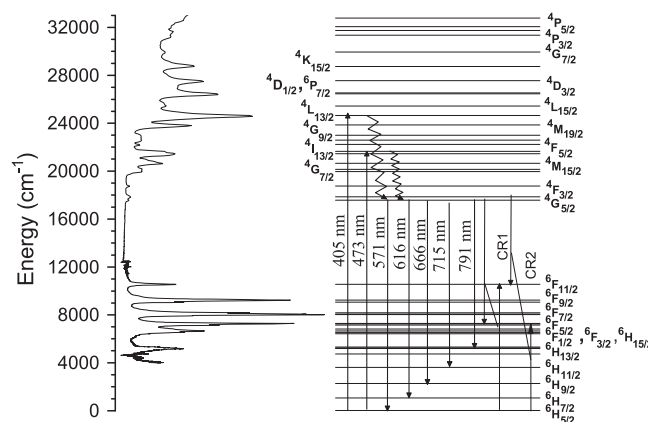
The structural parameters such as lattice parameter (Å), crystallite size and lattice strain are reported in Table 1 for Sm^{3+} -doped LuGG nano-garnets. It is found that the crystallite size obtained using Scherrer's equation is smaller than that computed by W–H method. This is due to the fact that the Scherrer's equation could not take into the account of lattice strain on the peak broadening. Similar observations have been reported elsewhere [20,21] and are shown in Table 1. It can also be seen from Table 1 that the strain is increased with increase of Sm^{3+} ion concentration due to substitution of lower ionic radii Lu^{3+} ions (1.001 Å) in dodecahedral sites in garnet structure by higher ionic radius of Sm^{3+} ion (1.079 Å). Moreover, higher doping concentration would give rise to defects, results in the distorted local structure. Similar results have also been observed from Eu^{3+} -doped LaOCl nano-crystals [22].

The size distribution of the LuGG1Sm nano-garnet was measured by DLS technique and shown in Fig. 2. The average diameter of the particle (d_p) can be calculated from the diffusion coefficient D using the Stokes–Einstein equation [23],

$$d_p = \frac{kT}{3\pi\mu D} \quad (2)$$

where k is the Boltzmann constant (in J K^{-1}); T is the absolute temperature (in K); and μ is the viscosity of the medium (in $\text{kg m}^{-1} \text{s}^{-1}$). The distribution of the nano-garnets in different sizes ranging from 20 to 55 nm with a maximum at 33 nm has been observed and is in good agreement with XRD results. As can be seen from Fig. 2, the maximum number of nano-garnets has been grown to about 33 nm in size and similar results have been noticed for all the Sm^{3+} -doped LuGG nano-garnets.

The diffuse reflectance spectrum of LuGG10Sm nano-garnet is shown in Fig. 3. As can be seen, the absorption bands appeared in the UV-visible-NIR region and their positions are similar to Sm^{3+} :

**Fig. 2.** Particle size distribution histogram of 1.0 mol% Sm^{3+} -doped LuGG nano-garnets.**Fig. 3.** Diffuse reflectance spectrum of 10.0 mol % Sm^{3+} -doped LuGG nano-garnets along with its partial energy level structure. CR1 and CR2 are cross relaxation channels.

LaCl_3 [24]. The spectrum consists of 22 sharp peaks due to $4f^5 \rightarrow 4f^5$ transitions, which are centered at 5200, 6658, 7278, 8019, 9233, 10,570, 17,699, 18,761, 20,080, 20,618, 21,413, 22,173, 22,727, 23,809, 24,570, 25,445, 26,385, 27,548, 28,735, 30,030, 31,545, and $32,573 \text{ cm}^{-1}$ corresponding to the ${}^6\text{H}_{5/2} \rightarrow {}^6\text{H}_{13/2}$, (${}^6\text{F}_{1/2}$, ${}^6\text{F}_{3/2}$, ${}^6\text{H}_{15/2}$), ${}^6\text{F}_{5/2}$, ${}^6\text{F}_{7/2}$, ${}^6\text{F}_{9/2}$, ${}^6\text{F}_{11/2}$, ${}^4\text{G}_{5/2}$, ${}^4\text{F}_{3/2}$, ${}^4\text{G}_{7/2}$, ${}^4\text{M}_{15/2}$, ${}^4\text{I}_{13/2}$, ${}^4\text{F}_{5/2}$, ${}^4\text{G}_{9/2}$, ${}^4\text{M}_{19/2}$, ${}^4\text{L}_{13/2}$, (${}^4\text{G}_{11/2}$, ${}^4\text{L}_{15/2}$), (${}^4\text{D}_{1/2}$, ${}^6\text{P}_{7/2}$), ${}^4\text{D}_{3/2}$, ${}^4\text{K}_{15/2}$, ${}^4\text{G}_{7/2}$, ${}^4\text{P}_{3/2}$, and ${}^4\text{P}_{5/2}$ transitions, respectively. The most intense line at 407 nm ($24,570 \text{ cm}^{-1}$) is assigned to spin-allowed ${}^6\text{H}_{5/2} \rightarrow {}^4\text{L}_{13/2}$ transition which is appropriate for optical pumping. The dense energy level

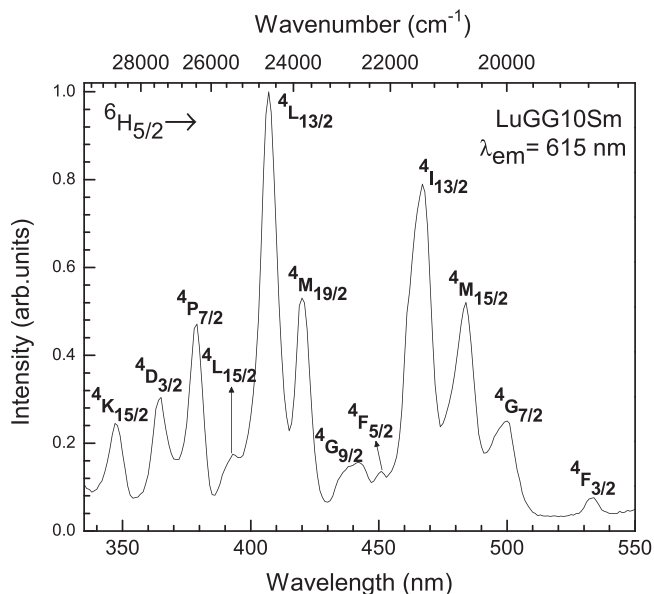


Fig. 4. Excitation spectrum of 10.0 mol% Sm^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ nano-garnet.

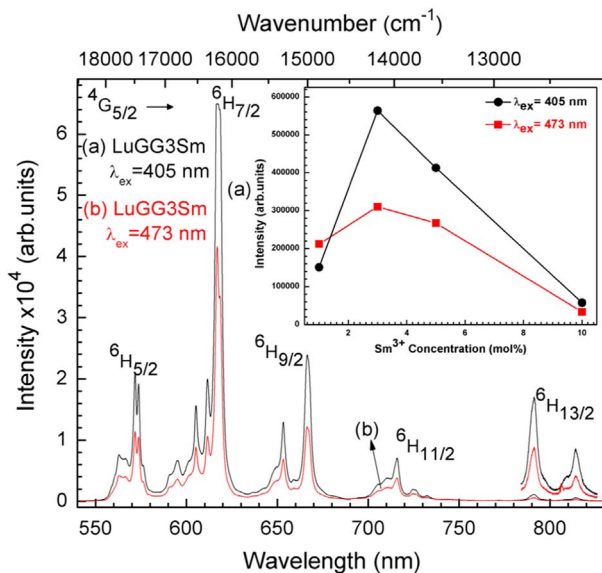


Fig. 5. Emission spectra of 3.0 mol% Sm^{3+} -doped LuGG nano-garnets under (a) $\lambda_{\text{ex}}=405$ nm and (b) $\lambda_{\text{ex}}=473$ nm. Inset shows the emission intensities of Sm^{3+} -doped LuGG nano-garnets under 405 (●) and 473 nm (■) excitations.

diagram favors fast population of the metastable level $^4\text{G}_{5/2}$ by non-radiative multiphonon relaxation processes from the upper levels. The excitation spectrum of 10.0 mol% of Sm^{3+} -doped LuGG nano-garnets, measured by monitoring the emission at 615 nm, is shown in Fig. 4. The observed peaks are at about 348, 364, 379, 393, 407, 420, 440, 451, 467, 485, 500, and 533 nm, assigned to the transitions from the $^6\text{H}_{5/2}$ ground level to various excited levels of $^4\text{K}_{15/2}$, $^4\text{D}_{3/2}$, $^4\text{D}_{1/2}$, $^6\text{P}_{7/2}$, $^4\text{L}_{15/2}$, $^4\text{L}_{13/2}$, $^4\text{M}_{19/2}$, $^4\text{G}_{9/2}$, $^4\text{F}_{5/2}$, $^4\text{I}_{13/2}$, $^4\text{M}_{15/2}$, $^4\text{G}_{7/2}$, and $^4\text{F}_{3/2}$, respectively. The positions and relative intensities of the excitation peaks agree well with the absorption spectrum.

The luminescence spectra of Sm^{3+} -doped LuGG nano-garnets have been measured for Sm^{3+} concentrations under both 405 and 473 nm diode laser excitations at similar experimental conditions and are shown in Fig. 5 only for LuGG3Sm nano-garnets. As can be seen from Fig. 5, the emission spectra consist of different emissions at 571, 616, 666, 715, and 791 nm corresponding to the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{5/2}$, $^6\text{H}_{7/2}$, $^6\text{H}_{9/2}$, $^6\text{H}_{11/2}$, and $^6\text{H}_{13/2}$ transitions, respectively. It is observed

that no appreciable variations in the emission peak positions under 405 and 473 nm excitations are found except for luminescence intensity. Of these emissions, the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition (orange-reddish) exhibits higher intensity. The $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{5/2}$ and $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transitions are magnetic-dipole in nature according to the selection rule ($\Delta J=0, \pm 1$), while the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{9/2}$ and $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{11/2}$ transitions are electric-dipole in nature. The $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition is the most pronounced one, which finds application in orange-reddish emitting material. The relative intensity ratio of electric-to-magnetic dipole transitions can be used to know the local symmetry around the RE^{3+} ions. The smaller value of the ratio indicates that there is no distortion from the inversion symmetry [15]. The ratio ($^4\text{G}_{5/2} \rightarrow ^6\text{H}_{9/2}/^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$) is found to be about 0.4 for all Sm^{3+} -doped LuGG nano-garnets. Hence, the Sm^{3+} ions are embedded in dodecahedral sites without distortion of the LuGG structure. The experimental branching ratios, β_{exp} for various $^4\text{G}_{5/2} \rightarrow ^6\text{H}_J$ ($J=5/2-13/2$) luminescent transitions have been evaluated and are found to be 17%, 54%, 20%, 7%, and 2%, respectively, for LuGG1Sm nano-garnet. It was established that the transition with β_{exp} value nearly equal to 50%, i.e. the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ emission, could be considered as a potential laser transition.

The intensities of $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition are evaluated for LuGG1Sm, LuGG3Sm, LuGG5Sm and LuGG10Sm samples under 405 and 473 nm excitations and are shown in inset of Fig. 5. It is observed that intensity is larger for 405 nm excitation due to higher absorbance for Sm^{3+} ions at 405 nm corresponds to $^6\text{H}_{5/2} \rightarrow ^4\text{L}_{13/2}$ transition than that at 473 nm corresponds to $^6\text{H}_{5/2} \rightarrow ^4\text{I}_{13/2}$ transition (see Fig. 3). It is also observed that the intensity increased up to 3.0 mol% and then decreased, due to the presence of energy transfer between Sm^{3+} ions and/or defect sites on the surface of the nano-garnets that result in concentration quenching. This indicates that the optimum molar concentration of Sm^{3+} ions is about 3.0 mol%.

The prospect of energy transfer between activators is inversely proportional to the n^{th} power of the distance between the activator ions [25]. Thus, the probabilities of energy migration in Sm^{3+} ions increase with increase in concentration as the distance between the Sm^{3+} ions decreases. Blasse suggested that the critical energy transfer distance (R_c) can be calculated by using the following equation [26]:

$$R_c = 2[3V/4\pi x_c N]^{1/3} \quad (3)$$

where V is the volume of the unit cell; x_c is the optimal concentration; and N gives cations number in the unit cell and it is 24 for LuGG nano-garnets. By substituting the values of V , x_c , and N as $1810 \text{ \AA}^3$, 0.03, and 24 [27], respectively, the R_c is estimated to be $16.8 \text{ \AA}$. Similar observations have been noticed for 2.0 mol% Sm^{3+} ions in CaWO_4 nanophosphor ($R_c=19.62 \text{ \AA}$) [28] and 4.0 mol% Sm^{3+} ions in Gd_2MoO_6 phosphor ($R_c=14.17 \text{ \AA}$) [29].

Further, energy transfer among RE^{3+} ions may be described by three different methods such as exchange interaction, radiative transfer and multipolar interaction [30]. The obtained R_c indicates that there is no possibility for exchange interaction since this interaction is predominant only for smaller separations ($R_c \leq 6 \text{ \AA}$) [31]. Generally, the radiation re-absorption occurs when there is an overlap between emission and absorption spectra of the ions. Therefore, this mechanism also does not intend to occur in this case. As a result, the multipolar interactions are assumed to be the only responsible for the energy transfer mechanism.

In order to analyze the multipolar interactions between the Sm^{3+} ions in the LuGG nano-garnet, the luminescence decay curves for the $^4\text{G}_{5/2}$ level for various Sm^{3+} ion concentrations have been measured and are shown in Fig. 6. All the decay curves exhibit non-exponential nature, as expected, due to energy transfer between Sm^{3+} ions. Hence, the effective lifetime τ_{eff} can

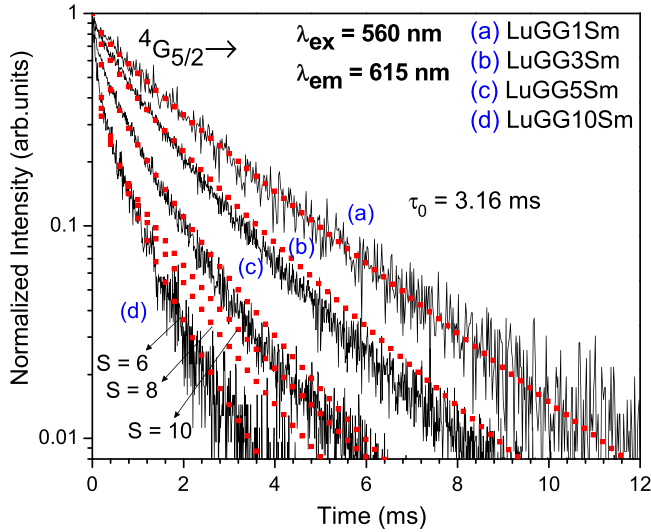


Fig. 6. Decay curves for the $^4G_{5/2}$ level of Sm^{3+} :LuGG nano-garnets. The best IH fit for $S=6$ is shown for all Sm^{3+} :LuGG nano-garnets and $S=8$ and 10 is shown for 10.0 mol% Sm^{3+} -doped LuGG nano-garnets.

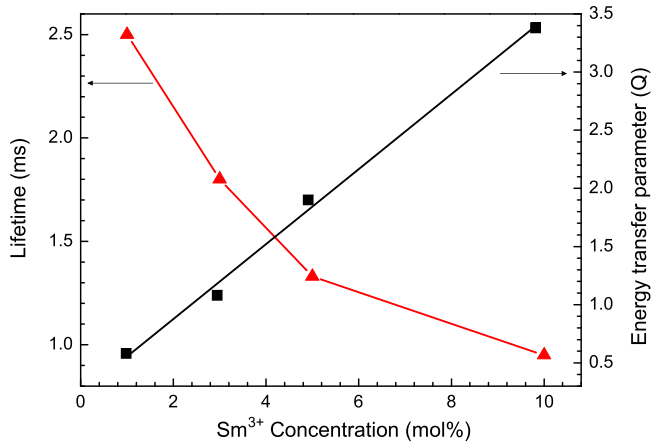


Fig. 7. Variation of lifetime (▲) and energy transfer parameter (■) as a function of Sm^{3+} concentration in LuGG nano-garnets.

Table 2

Lifetime of $^4G_{5/2}$ level of Sm^{3+} ions in different crystalline materials and their phonon energy.

S.No	Sample	Lifetime (ms)	Phonon energy (cm^{-1})	Ref.
1	LuGG:1.0 mol% Sm^{3+} (nano)	2.50	766	Present
2	LuGG:3.0 mol% Sm^{3+} (nano)	1.80		Present
3	LuGG:5.0 mol% Sm^{3+} (nano)	1.33		Present
4	LuGG:10.0 mol% Sm^{3+} (nano)	0.95		Present
5	$CaWO_4$:2.0 mol% Sm^{3+} (nano)	0.44	912	28
6	Gd_2MoO_6 :0.5 mol% Sm^{3+} (bulk)	0.65		29
7	Gd_2MoO_6 :1.0 mol% Sm^{3+} (nano)	0.58	853	29
8	Gd_2MoO_6 :2.0 mol% Sm^{3+} (bulk)	0.47		29
9	Gd_2MoO_6 :4.0 mol% Sm^{3+} (bulk)	0.29		29
10	$LiYF_4$:1.0 mol% Sm^{3+} (bulk)	4.8		34
11	$LiYF_4$:5.0 mol% Sm^{3+} (bulk)	1.5	446	34
12	$LiYF_4$:10.0 mol% Sm^{3+} (bulk)	0.6		34
13	KY_3F_{10} :1.0 mol% Sm^{3+} (bulk)	4.36	495	36

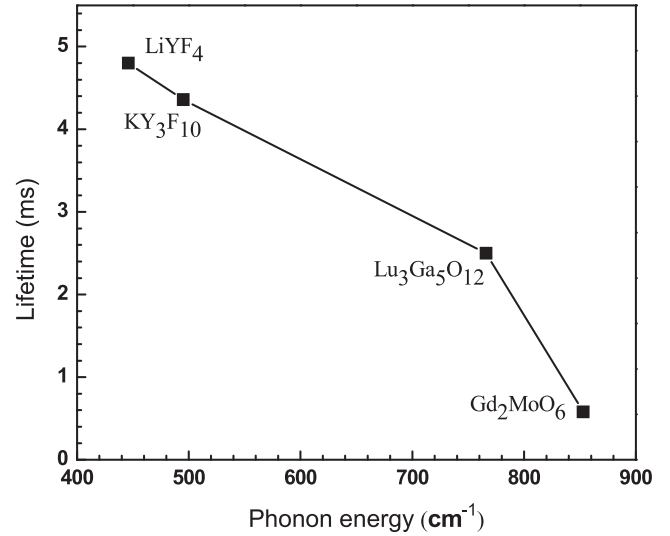


Fig. 8. Variation of lifetime of $^4G_{5/2}$ level of 1.0 mol% of Sm^{3+} -doped crystalline materials as a function of phonon energy.

be evaluated by using the equation

$$\tau_{\text{eff}} = \frac{\int tI(t)dt}{\int I(t)dt} \quad (4)$$

where $I(t)$ is the intensity of the luminescence at time t . The lifetime of the $^4G_{5/2}$ level of Sm^{3+} ions was found to be 2.5, 1.8, 1.33 and 0.95 ms for 1.0, 3.0, 5.0 and 10.0 mol% of Sm^{3+} -doped LuGG nano-garnets, respectively (see Fig. 7). The observed shortening of lifetime with the increase of Sm^{3+} ion concentration is due to non-radiative energy transfer through ($^4G_{5/2}; ^6H_{5/2}$) $\rightarrow$ ($^6F_{5/2}; ^6F_{11/2}$) and ($^4G_{5/2}; ^6H_{5/2}$) $\rightarrow$ ($^6F_{11/2}; ^6F_{5/2}$) cross-relaxation channels, labeled as CR1 and CR2 in Fig. 3. The obtained lifetime values are compared with the lifetime values of Sm^{3+} -doped $CaWO_4$ [28] nanocrystals and Gd_2MoO_6 [29], $LiYF_4$ [34], and KY_3F_{10} [36] bulk materials and are shown in Table 2. Similar concentration (1.0 mol %) of Sm^{3+} ions in different hosts have been chosen to plot the variation of lifetimes of $^4G_{5/2}$ level as function of phonon energy. It is observed that the lifetime of $^4G_{5/2}$ level of Sm^{3+} ion decreases as the phonon energy of the host [28,32,33,35,37] increases, as shown in Fig. 8. It is also found from the Table 2 that the lifetime of $^4G_{5/2}$ level of Sm^{3+} -doped LuGG nano-garnets is high compared to other oxide materials i.e Sm^{3+} -doped Gd_2MoO_6 [28] and $CaWO_4$ [29] phosphors and low compared to fluoride materials [34,36].

In order to know the dominant mechanism of interaction between the Sm^{3+} ions, the non-exponential decay curves have been analyzed by Inokuti and Hirayama model [38]. According to this model, luminescence intensity, $I(t)$, is described as,

$$I(t) = I_0 \exp \left[-\left(\frac{t}{\tau_0} \right) - Q \left(\frac{t}{\tau_0} \right)^{\frac{3}{S}} \right] \quad (5)$$

where I_0 is the emission intensity at $t=0$; $I(t)$ is the emission intensity after excitation; τ_0 is the intrinsic lifetime; $S=6, 8$ and 10 indicates the multipolar interaction mechanism for dipole–dipole, dipole–quadrupole and quadrupole–quadrupole interactions between the ions, respectively; and Q is the energy transfer parameter and is described by

$$Q = \frac{4\pi}{3} \Gamma [1 - 3/S] N_0 R_0^3 \quad (6)$$

where N_0 is the concentration of acceptors, which is approximately equal to the total concentration of Sm^{3+} ions; and R_0 is the critical transfer distance defined as a donor–acceptor separation for which the rate of energy transfer to the acceptors is equal to the rate of intrinsic decay of the donor.

All the decay curves are well fitted to Eq. (5) for $S=6$ with an intrinsic lifetime $\tau_0=3.16$ ms, indicating that the dominant mechanism of interaction between Sm^{3+} ions is of dipole–dipole type. The energy transfer parameter Q , obtained from the fitting, increases linearly with increase in Sm^{3+} ion concentration in accordance with Eq. (6), as shown in Fig. 7.

4. Conclusions

Nanocrystalline $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ garnets doped with various concentrations of Sm^{3+} ions have been prepared in single phase of cubic structure. The crystallite size estimated from both Debye–Scherrer and Williamson–Hall methods is in good agreement with dynamic light scattering technique. The nano-garnets of the present study emit intense orange–reddish emission corresponding to the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition under 405 nm diode laser excitation. The variation of the emission intensity with the Sm^{3+} concentration indicates that optimum concentration of Sm^{3+} ions in $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ nano-garnet is about 3.0 mol%, with a critical energy transfer distance of 16.8 Å. The decay curve analysis using Inokuti–Hirayama model reveals that dipole–dipole interaction between Sm^{3+} ions is responsible for non-exponential nature of the decay curves and quenching of lifetime. The present results indicate that the nano-garnets could be useful for the development of white light emitting devices.

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